

Kelvin Chen

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTACT INFORMATION | Medical Research Bldg 4<br>409 Lane Rd<br>Charlottesville, VA 22903                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Phone: (xxx) xxx-xxxx<br>Email: <a href="mailto:ddw4hp@virginia.edu">ddw4hp@virginia.edu</a><br>Homepage: <a href="https://k9chen.github.io">https://k9chen.github.io</a>                                                                                      |
| RESEARCH INTERESTS  | Translational and clinical neuroscience, neuroimmunology, cerebrovascular physiology, neuropathology, neuropharmacology, neurosurgery                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                |
| EDUCATION           | <b>University of Virginia</b> , Charlottesville, VA<br>BA, Neuroscience; Minor, Bioethics<br>• Advisor(s): <a href="#">Jianjie Ma, PhD</a> ; <a href="#">Ling Qi, PhD</a><br>• Thesis: <i>MG53/CitH3 modulation attenuates NETosis-mediated neuroinflammation</i><br><b>Virginia Peninsula Community College</b> , Hampton, VA<br>Dual Enrollment, Biological Science & Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Aug 2022 – May 2026<br><br><br><br><br>Sept 2020 – Jun 2022                                                                                                                                                                                                    |
| RESEARCH EXPERIENCE | <b>UVA School of Medicine</b> , Charlottesville, VA<br><i>Undergraduate Researcher, Dept. of Surgery</i><br>Advisor: <a href="#">Jianjie Ma, PhD</a><br>• Treatment of cerebral ischemia with CHO cell-derived rhMG53<br><i>Undergraduate Researcher, Dept. of Neurosurgery</i><br>Advisor: <a href="#">Petr Tvrdik, PhD</a><br>• Recombinase-based spatiotemporal mapping of myeloid cells<br><i>Undergraduate Researcher, Dept. of Neuroscience</i><br>Advisor: <a href="#">Lulu Jiang, MD, PhD</a><br>• Proximity labeling of translocated RNA with TurboID in iPSCs<br><i>Undergraduate Researcher, Dept. of Pharmacology</i><br>Advisor: <a href="#">Julius Zhu, PhD</a><br>• Image analysis with genetically encoded fluorescent sensors<br><b>Barrow Neurological Institute</b> , Phoenix, AZ<br><i>Research Intern, Dept. of Translational Neuroscience</i><br>Advisor: <a href="#">Richard Dortch, PhD</a><br>• Numerical simulation of multi-compartmental dMRI models<br><i>Research Intern, Dept. of Translational Neuroscience</i><br>Advisor: <a href="#">Richard Dortch, PhD</a><br>• Automatic segmentation of peripheral nerve histology images<br><b>Hampton University</b> , Hampton, VA<br><i>Research Intern, Dept. of Chemistry &amp; Biochemistry</i><br>Advisor: <a href="#">Peter Njoki, PhD</a><br>• Viral kinetic ODE-based modeling of antigen-conjugated NPs | Dec 2024 – Present<br><br><br><br><br>Jan 2024 – Dec 2024<br><br><br><br><br>Aug 2023 – Jan 2024<br><br><br><br><br>Sept 2022 – May 2023<br><br><br><br><br>May 2024 – Aug 2024<br><br><br><br><br>May 2023 – Aug 2023<br><br><br><br><br>Sept 2021 – May 2022 |
| AWARDS & GRANTS     | Ingrassia Family Research Award (\$2,000), UVA<br>Harrison Undergraduate Research Award (\$5,000), UVA<br>Echols Scholarship (top 5% of class), UVA<br>Dean’s List of Distinguished Students, UVA<br>University Achievement Award Scholarship (full-tuition), UVA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025<br>2025<br>2023<br>2023 – 2025<br>2022 – 2026                                                                                                                                                                                                             |

PUBLICATIONS &  
PRESENTATIONS

 GOOGLE SCHOLAR

- [C.1] Ketsiri, T., **Chen, K.**, Xu, J., Dortch, R. D. *Numerical validation of multi-compartment diffusion biomarkers of peripheral nerve trauma*. 33rd Annual Meeting & Exhibition of the International Society for Magnetic Resonance in Medicine, Honolulu, HI, May 2025. [[Abstract](#)]
- [C.2] Ketsiri, T., **Chen, K.**, Xu, J., Dortch, R. D. *Validation of multi-compartmental diffusion MRI models for peripheral nerve trauma*. Gordon Research Conference on In Vivo Magnetic Resonance, Andover, NH, Jul 2024. [[Abstract](#)]
- [O.1] **Chen, K.**, Lee, K. E., Park, K. H., Ma, J. *MG53/CitH3 modulation attenuates NETosis-mediated neuroinflammation*. UVA Fozdar Symposium for Undergraduate Neuroscience Research, Charlottesville, VA, Apr 2026. [[Slides](#)]
- [P.1] **Chen, K.**, Lee, K. E., Park, K. H., Ma, J. *Co-targeting MG53 and CitH3 attenuates cerebral ischemia-reperfusion injury*. UVA Undergraduate Research Symposium, Charlottesville, VA, Apr 2026. [[Poster](#)]
- [P.2] **Chen, K.**, Lee, K. E., Park, K. H., Ma, J. *Co-targeting MG53 and CitH3 attenuates cerebral ischemia-reperfusion injury*. UVA Fozdar Poster Session, Charlottesville, VA, Dec 2025. [[Poster](#)]
- [P.3] **Chen, K.**, Ketsiri, T., Dortch, R. D. *Spherical mean diffusion-weighted MRI reveals peripheral axonal pathology following trauma*. BNI Undergraduate Research Symposium, Phoenix, AZ, Aug 2024. [[Poster](#)]
- [P.4] **Chen, K.**, Sadrabadi, M. S., Dortch, R. D. *Geometry-informed multi-compartmental diffusion MRI modeling of sciatic nerve crush*. BNI Undergraduate Research Symposium, Phoenix, AZ, Aug 2023. [[Poster](#)]

TEACHING  
EXPERIENCE

**University of Virginia**, Charlottesville, VA

*Teaching Assistant, Dept. of Chemistry*

CHEM 1810 Principles of Chemical Structure (Accelerated) F24, F25

CHEM 2321 Organic Chemistry II Laboratory S24, S25

CHEM 2311 Organic Chemistry I Laboratory F23, F24, F25

CHEM 1811 Principles of Chemical Structure Laboratory (Accelerated) F23

*Teaching Assistant, Dept. of Physics*

PHYS 2030 Principles of Physics I Laboratory (Pre-Health) F25

PHYS 1429 Introductory Physics I Laboratory (Engineering) S25

*Teaching Assistant, Dept. of Psychology*

PSYC 3210 Research Methods: Psychobiology Laboratory S24

SERVICE &  
OUTREACH

**UVA Office of Citizen Scholar Development**, Charlottesville, VA

*Symposium Volunteer*

Apr 2023, Apr 2024

- UVA Undergraduate Research Symposium

**UVA School of Medicine**, Charlottesville, VA

*Symposium Volunteer*

Mar 2025

- Advancing Innovations in Translational Research into Human Applications Symposium in Honor of Hiroshi Takeshima

**W. M. Keck Center for Cellular Imaging**, Charlottesville, VA

*Workshop Volunteer*

Mar 2024

- 21st Annual Workshop on FLIM and FRET Microscopy and Metabolic Imaging

|              |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OTHER        | Outreach Ministry Volunteer, Grace Christian Fellowship                                                                                                                                                                           | 2023 – Present                                                                                                                                                                                                                                                       |
|              | Senior Associate, Blossom Together                                                                                                                                                                                                | 2023 – Present                                                                                                                                                                                                                                                       |
|              | Editor, Grounds: The Virginia Journal of Bioethics                                                                                                                                                                                | 2023 – Present                                                                                                                                                                                                                                                       |
|              | Senior Translator, The Cavalier Daily                                                                                                                                                                                             | 2023 – Present                                                                                                                                                                                                                                                       |
|              | Editor, Oculus: The Virginia Journal of Undergraduate Research                                                                                                                                                                    | 2022 – Present                                                                                                                                                                                                                                                       |
|              | Investigator, University Judiciary Committee                                                                                                                                                                                      | 2022 – Present                                                                                                                                                                                                                                                       |
|              | Health & Medical Services Volunteer, Madison House                                                                                                                                                                                | 2022 – 2023                                                                                                                                                                                                                                                          |
| AFFILIATIONS | American Neurological Association                                                                                                                                                                                                 | Jun 2024 – Present                                                                                                                                                                                                                                                   |
| SKILLS       | Programming: Python, MATLAB, R, C/C++, HTML/CSS, JavaScript, L <sup>A</sup> T <sub>E</sub> X<br>Frameworks: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV<br>Languages: English, Mandarin                              |                                                                                                                                                                                                                                                                      |
| REFERENCES   | <b>Jianjie Ma, PhD</b><br>Professor & Director<br>Division of Surgical Sciences<br>Dept. of Surgery<br>UVA School of Medicine<br><a href="mailto:qga8zx@virginia.edu">qga8zx@virginia.edu</a>                                     | <b>Richard Dortch, PhD</b><br>Associate Professor, Director, & Chair<br>Neuroimaging Innovation Center<br>Dept. of Translational Neuroscience<br>Barrow Neurological Institute<br><a href="mailto:richard.dortch@barrowneuro.org">richard.dortch@barrowneuro.org</a> |
|              | <b>Ammasi Periasamy, PhD</b><br>Professor, Director, & Founder<br>W. M. Keck Center for Cellular Imaging<br>Dept. of Biology & Biomed. Eng.<br>University of Virginia<br><a href="mailto:ap3t@virginia.edu">ap3t@virginia.edu</a> | <b>Andreas Gahlmann, PhD</b><br>Associate Professor<br>Center for Membrane & Cell Physiology<br>Dept. of Mol. Physiol. & Biol. Phys.<br>UVA School of Medicine<br><a href="mailto:ag5vu@virginia.edu">ag5vu@virginia.edu</a>                                         |